

MacSync CX1 4G LTE Cellular RS485 & Analog Data Logger Datasheet

Product Overview

MacSync CX1 is an industrial-grade **4G LTE Cellular Data Logger** designed to connect **RS485 (Modbus RTU)** and **analog sensors (4–20 mA / 0–10 V)** directly to the cloud for real-time monitoring and analytics. Built for demanding outdoor deployments, CX1 features a rugged enclosure, stable performance, and reliable cellular connectivity — making it ideal for remote industrial sites where Wi-Fi or Ethernet is not available.

Powered by a wide **9–36 V DC input**, MacSync CX1 is engineered for permanent installations in utilities, factories, water infrastructure, energy metering, and process monitoring systems. It supports multi-channel data acquisition, periodic and event-based reporting, and enables instant alerts for critical parameter changes.

With an **in-house high-efficiency internal antenna** and optimized low-noise RF design, CX1 ensures stable long-range connectivity in harsh environments. Whether you are monitoring pumps, pressure transmitters, flow meters, energy meters, or industrial machines — **MacSync CX1 transforms field signals into cloud-ready data with industrial reliability.**





Key Features

4G LTE Cellular Data Logger

Direct-to-cloud connectivity for remote monitoring without Wi-Fi, Ethernet, or gateway dependency.

Analog Inputs (4–20 mA / 0–10 V)

Supports industry-standard analog sensors such as pressure, level, flow, and temperature transmitters.

Outdoor Rugged Enclosure

Built for harsh field environments with reliable performance in dust, heat, and moisture conditions.

Smart Reporting & Alerts

Supports periodic reporting and threshold-based alerts for critical parameter changes.

Local BLE Configuration

Quick on-site setup and diagnostics using the Macnman Maya mobile app.

RS485 (Modbus RTU) Integration

Seamlessly connects with industrial PLCs, energy meters, flow meters, and smart instruments.

Wide Input Power (9–36 V DC)

Designed for stable outdoor and industrial installations with wide-range DC power compatibility.

In-House High-Efficiency Internal Antenna

Optimized RF design ensures stable long-range cellular connectivity for remote deployments.

Off-the-Shelf Cloud Compatibility

Works with AWS IoT Core, Azure IoT Hub, ThingsBoard, Ubidots, Datacake, TagoIO, Blynk, and custom cloud.

Secure & Reliable Data Delivery

Supports encrypted communication and robust buffering strategies for dependable industrial telemetry.

Suited Applications

Smart Water Utilities

Continuous level monitoring for underground water tanks, overhead tanks, reservoirs, and sumps.

Manhole & Sewer Level Monitoring

Detect overflow risks, blockages, and sewer flooding in drainage networks.

Flood & Stormwater Monitoring

Real-time alerts for rising levels in rivers, canals, storm drains, and flood-prone zones.

Mining & Quarry Operations

Level tracking in slurry pits, water sumps, and tailing dams for safety assurance.

Oil & Fuel Storage Tanks

Level measurement for diesel storage, fuel stations, and industrial oil chambers.

Environmental Risk Monitoring

Early warning for spill detections, groundwater rise, or hazardous waste overflow.

Industrial Wastewater Management

Accurate level detection in effluent treatment plants, sludge pits, and chemical tanks.

Smart City Infrastructure

Remote monitoring of drainage networks to prevent urban flooding and road closures.

Remote & Unmanned Locations

Low-maintenance, long-range LoRaWAN® monitoring where wired installation is impractical.

Technical Specifications

General Specifications

Part Order Number	MacSync-CX1
Product Brand	Macnman Technologies Pvt Ltd

Interfaces Available

RS485 Interface	1 (Switchable)
4-20 mA Analog Channels	x 2 (Switchable)
0-10 V Analog Channels	x 2 (Switchable)
Digital Inputs	x 2 (Switchable)
ADC Resolution	12-bit

 Note

The device use same channels for the RS485 & Analog inputs, either can be used at a time.

Wireless Transmission

Communication Module	Quectel EC200 Series
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Network Technology	4G LTE (Cat 1) 2G GSM (Fallback)
Supported LTE Bands*	B1 / B3 / B5 / B8 / B40 / B41 (Region dependent)
Supported GSM Bands*	900 / 1800 MHz
Tx Power	LTE: Up to +23 dBm GSM: Up to +33 dBm
Receiver Sensitivity*	LTE: Up to -100 dBm (typical) GSM: Up to -109 dBm (typical)
Cloud Protocols	MQTT / HTTPS
Antenna	External high-efficiency cellular antenna
SIM Support	Nano SIM
Data Transmission	Configurable reporting interval with event-based alerts

LED Status

LED Type	1 Internal Multi-Color LED
Solid Blue	Device is connected with Maya App
Flashing Blue	OTA (Over-the-Air) update in progress
Slow Blinking Green	Attempting to connect to the network
Single Red Blink	A confirmed parameter of MacSync is being modified

Device Configuration

Configuration	Via Macnman Maya Android App
Uplink Modes	Time Based / Threshold Based / Sampling Based
Stored Records*	30,000 Records
Remote Configuration	JSON Format Data - from server

Supported Cloud Platforms

Supports Major Cloud Platforms	AWS IoT Core Microsoft Azure IoT Hub ThingsBoard Ubidots Datacake
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TagoIO
Blynk
Custom Cloud via MQTT / HTTPS

Physical Characteristics

Power Supply	9-36 V DC
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)
Ingress Protection	IP65 standard (IP67 optional variant)
Housing Material	Glass filled Nylon
Color	Matte White
Weight	~110g
Mounting Options	Pole Mounting / Wall Mounting

Electrical Characteristics

Input Voltage Range	9-36 V DC (External Supply)
Power Consumption	~100 µA in sleep mode ~50 to 250 mA during transmission
Input Impedance	>10 MΩ for voltage inputs

Communication Options

BLE Connectivity	For local configuration and diagnostics via Maya App
Cloud Protocols	MQTT / HTTPS
Serial Interface	RS485 Modbus RTU support (optional)

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

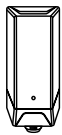
Data Encryption	AES-128 end-to-end encryption
Authentication	OTAA activation modes
Firmware Protection	Secure boot and signed OTA updates

MacSync CX1 Variants

VARIANT NAME	POWER SOURCE	INPUT VOLTAGE	SOLAR SUPPORT	EXTERNAL FLASH	IDEAL FOR
MacSync CX1 - PO	External Power	9–36 V DC			Outdoor industrial installations with always-on cellular connectivity

What's in the Box?

Make sure box contains following items before installations



MacSync



Mounting
Clip



User manual &
Warranty Card



Junction
Box*



Mounting
Screws

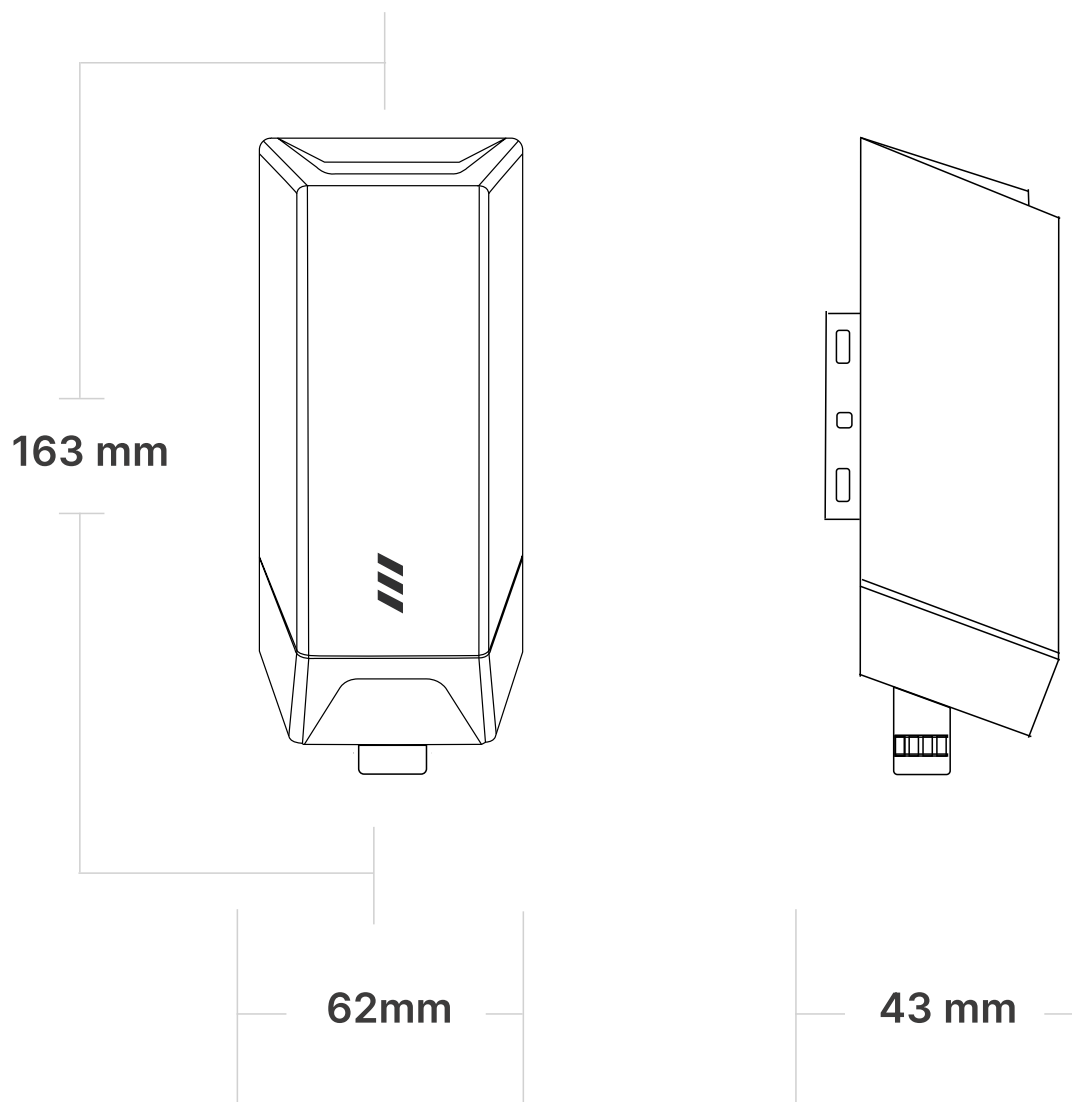


Power
Adapters*

Note

Accessories must be purchased separately & may not be included in the box.

MacSync CX1 Dimensions



Frequently Asked Questions

What is the MacSync CX1 used for?

MacSync CX1 is an industrial 4G LTE cellular data logger connecting RS485 Modbus RTU and 4-20 mA / 0-10 V analog sensors to the cloud. Typical applications include Smart Water Utilities, Manhole & Sewer Level Monitoring, Flood & Stormwater Monitoring, Industrial Wastewater Management, Mining & Quarry Operations and Smart City Infrastructure.

What inputs and outputs does the MacSync CX1 have?

The MacSync CX1 provides the following interfaces:

- RS485 Interface: 1 (Switchable)
- 4-20 mA Analog Channels: x 2 (Switchable)
- 0-10 V Analog Channels: x 2 (Switchable)
- Digital Inputs: x 2 (Switchable)
- ADC Resolution: 12-bit

Which cellular networks does the MacSync CX1 support?

The MacSync CX1 uses a Quectel EC200 Series module for 4G LTE (Cat 1), 2G GSM (Fallback) connectivity. Supported LTE bands: B1 / B3 / B5 / B8 / B40 / B41 (Region dependent). Supported GSM bands: 900 / 1800 MHz. It takes a Nano SIM and sends data over MQTT / HTTPS.

How is the MacSync CX1 powered?

The MacSync CX1 is powered from 9-36 V DC (External Supply).

What is the IP rating and operating temperature of the MacSync CX1?

The MacSync CX1 has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

How is the MacSync CX1 configured?

The MacSync CX1 is configured with the Macnman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

Related Products and Resources

- Related datasheets: [MacSync LX1 RS485 Modbus RTU and analog to LoRaWAN datalogger](#), [MacEdge CX08 industrial edge gateway](#), [MacEdge CX10 industrial edge gateway](#), [MacSet CX1 4G LTE cellular relay controller](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)